02 – Cybersecurity Career Roles

1. Introduction

Cybersecurity is the practice of protecting computers, networks, applications and data from cyberattacks, unauthorized access and damage.

There are different roles in cybersecurity. Three important roles are:

1. SOC Analyst
2. VAPT / Penetration Tester
3. Incident Responder

Each role has a different responsibility, but all three help organizations improve their security.

2. SOC Analyst

A Security Operations Center (SOC) Analyst monitors an organization's computers, networks and systems to identify suspicious activities and possible cyberattacks.

Main Goal

The main goal of a SOC Analyst is to detect security threats as early as possible and help protect the organization's systems and data.

Daily Work

- Monitor security alerts and system logs.
- Identify suspicious login attempts and unusual network traffic.
- Investigate malware alerts and phishing emails.
- Analyze logs from firewalls, endpoints and servers.
- Use SIEM tools to collect and analyze security events.
- Escalate serious incidents to the incident response team.
- Document findings and prepare security reports.

Common Tools

Tool	Purpose
Splunk	Collects and analyzes logs using SIEM capabilities.
Microsoft Sentinel	Cloud-based SIEM and security analytics.
Wazuh	Security monitoring, log analysis and endpoint detection capabilities.
Wireshark	Captures and analyzes network traffic.
Microsoft Defender	Helps detect and investigate security threats.

Example

A SOC Analyst receives an alert showing multiple failed login attempts followed by a successful login from an unusual location. The analyst investigates the logs to determine whether the account may be compromised.

Team Type

SOC Analysts mainly work on the Blue Team, which focuses on defending systems and detecting threats.

3. VAPT / Penetration Tester

VAPT stands for Vulnerability Assessment and Penetration Testing.

- Vulnerability Assessment: Identifies and evaluates security weaknesses in systems, applications and networks.
- Penetration Testing: Safely tests whether authorized security weaknesses can be exploited and what impact they could have.

A penetration tester acts like an ethical hacker to find security weaknesses before malicious attackers can use them.

Main Goal

The main goal is to identify and validate security weaknesses and recommend fixes before attackers exploit them.

Daily Work

- Understand the authorized testing scope.
- Identify systems and applications included in the assessment.
- Scan for open ports and services.
- Find outdated software and security misconfigurations.
- Test web applications for vulnerabilities.
- Validate findings using safe, controlled methods.
- Assess the possible impact of vulnerabilities.
- Prepare a report containing evidence, severity and remediation steps.

Common Tools

Tool	Purpose
Nmap	Discovers hosts, open ports and services.
Burp Suite	Tests web application security.
OWASP ZAP	Helps identify web application vulnerabilities.
Nikto	Checks web servers for potentially dangerous configurations and files.
Kali Linux	A Linux distribution containing many security testing tools.
Nessus	Vulnerability scanning and assessment.

Example

A company authorizes a penetration tester to assess its website. The tester identifies an SQL injection vulnerability in a test environment, validates its impact without accessing unrelated data, and recommends parameterized queries to fix it.

Team Type

Penetration testers generally work on the Red Team or in an offensive security role. However, VAPT engagements are often separate from formal Red Team exercises.

4. Incident Responder

An Incident Responder investigates and manages cybersecurity incidents such as ransomware infections, compromised accounts, malware attacks and unauthorized access.

Main Goal

The main goal is to contain the incident, reduce damage, investigate what happened and help restore normal operations safely.

Daily Work

- Receive and assess security incident reports.
- Determine the scope and severity of an incident.
- Collect and preserve relevant logs and digital evidence.
- Isolate affected endpoints when necessary.
- Investigate suspicious processes, files and network connections.
- Identify how an attacker entered the system.
- Remove the threat and support system recovery.
- Prepare incident reports and recommend preventive measures.

Common Tools

Tool	Purpose
Velociraptor	Endpoint visibility and digital investigation.
KAPE	Collects and processes Windows forensic artifacts.
FTK Imager	Acquires and examines forensic images and files.
Autopsy	Analyzes digital evidence.
Volatility 3	Analyzes memory captures for forensic evidence.
Microsoft Defender for Endpoint	Endpoint detection, investigation and response.

Example

An organization discovers that a computer is infected with ransomware. The incident responder helps isolate the affected computer, preserve evidence, investigate the infection, remove the threat and support safe recovery.

Team Type

Incident Responders generally work on the Blue Team because their focus is defending systems and responding to attacks.

5. Blue Team vs Red Team

Blue Team

The Blue Team focuses on defending an organization's systems, networks and information against cyberattacks.

Main responsibilities:

- Monitor security alerts.
- Detect suspicious activities.
- Investigate threats.
- Respond to security incidents.
- Improve security controls.

Common tools: SIEM platforms, EDR tools, firewalls and log analysis tools.

Red Team

The Red Team simulates authorized attackers to test an organization's defenses and identify weaknesses.

Main responsibilities:

- Perform authorized reconnaissance.
- Test security controls.
- Simulate realistic attack techniques.
- Identify paths that could lead to compromise.
- Report findings and recommend improvements.

Common tools: Nmap, Burp Suite, Metasploit and other authorized security testing tools.

Blue Team vs Red Team Comparison

Basis	**Blue Team**	**Red Team**
Main purpose	Defend systems	Test defenses through simulated attacks
Approach	Defensive	Offensive
Activities	Monitoring, detection and response	Reconnaissance, exploitation and attack simulation
Typical roles	SOC Analyst, Incident Responder	Penetration Tester, Red Team Operator
Common tools	SIEM, EDR, Wireshark	Nmap, Burp Suite, Metasploit
Success means	Threats are detected and contained	Security weaknesses are demonstrated and reported
Focus	Prevention, detection and recovery	Finding and validating weaknesses

Note: Penetration testing and Red Team exercises are related but not identical. A penetration test often focuses on finding vulnerabilities in a defined scope, while a Red Team exercise simulates an adversary's objectives and tests the organization's detection and response capabilities.

6. Comparison of the Three Cybersecurity Roles

Basis	SOC Analyst	VAPT / Penetration Tester	Incident Responder
Primary goal	Detect threats	Find and validate weaknesses	Contain and investigate incidents
Main focus	Security monitoring	Security testing	Incident investigation and recovery
Daily work	Analyze alerts and logs	Scan and test systems	Collect evidence and investigate attacks
Common tools	Splunk, Sentinel, Wazuh	Nmap, Burp Suite, OWASP ZAP	KAPE, FTK Imager, Volatility 3
Team association	Blue Team	Red Team / offensive security	Blue Team
Typical output	Alerts and investigation reports	Vulnerability assessment report	Incident report and recovery recommendations
Key skills	Log analysis, networking, threat detection	Web security, networking, vulnerability testing	Forensics, log analysis, incident handling

7. How These Roles Work Together

Consider an organization that experiences a phishing attack.

1. **SOC Analyst:** Detects suspicious email activity and unusual logins.
2. **Incident Responder:** Investigates the compromise, contains affected accounts and preserves evidence.
3. **VAPT / Penetration Tester:** Later tests relevant security controls within an authorized scope to identify weaknesses that need improvement.

These roles complement one another. The SOC Analyst helps detect attacks, the Incident Responder helps manage active incidents, and the Penetration Tester helps identify security weaknesses before they are exploited.

8. Important Points to Remember

- SOC Analyst = Detect
- VAPT / Penetration Tester = Test
- Incident Responder = Contain and investigate
- Blue Team = Defend
- Red Team = Simulate attacks

- SIEM stands for Security Information and Event Management.
- VAPT stands for Vulnerability Assessment and Penetration Testing.
- All penetration testing must have proper authorization and a clearly defined scope.

9. Conclusion

SOC Analysts, VAPT professionals and Incident Responders play important roles in cybersecurity. SOC Analysts monitor systems for suspicious activity, penetration testers identify security weaknesses, and incident responders investigate and manage cyberattacks. Working together, these professionals help organizations prevent attacks, detect threats and recover from security incidents.